

Calibrated Base Closure in a Primitive Möbius Tail: Ramanujan-Square Zero Modes and Passage Beyond the Poisson Length

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Abstract

Let $P_{m,S}$ be the calibrated residual prefix in a primitive opened row $m = \ell d$, where $\ell \asymp L$ is prime, d is squarefree, and $R \leq S$. Earlier exact-conductor estimates close the retained nonzero additive spectrum of the centered selected square when S is strictly below the Poisson length. They do not, by their displayed formulas, estimate the connected additive zero mode. This paper closes that missing base mode for the actual opened-row provenance and then restores the calibrated physical square.

On the primitive orbit $(j, H) = 1$, write

$$Z_{m,S}(j) = \sum_{\substack{u \leq S \\ (u, mH)=1}} b_R(u) \left(\mathbf{1}_{u|m j+h} - \frac{1}{u} \right), \quad P_{m,S} = Z_{m,S} + E_S(m).$$

For distinct rows m, n , the complete-period zero kernel of $Z_{m,S}Z_{n,S}$ is

$$K_S^0(m, n) = \sum_{u, v \leq S} \frac{b_R(u)b_R(v)}{uv} ((u, v)\mathbf{1}_{(u,v)|m-n} - 1),$$

with the primitive row-coprimality restrictions understood. The identity

$$g\mathbf{1}_{g|\Delta} - 1 = \sum_{\substack{k|g \\ k>1}} c_k(\Delta)$$

factorizes it exactly as

$$K_S^0(m, n) = \sum_{k>1} c_k(m-n) B_{k,S}(m) B_{k,S}(n), \quad B_{k,S}(m) = \sum_{\substack{u \leq S \\ k|u \\ (u, mH)=1}} \frac{b_R(u)}{u}.$$

For $k \leq X^\kappa$, an exact multiple-density identity for the finite Ramanujan model and the classical uniform Möbius calibration give

$$B_{k,S}(m) = \frac{\mu(k)}{\varphi(k)} \delta_H(m) + O_A \left(\frac{1}{\varphi(k)(\log X)^A} \right), \quad \delta_H(m) \ll \frac{\log \log(3X)}{L}.$$

For $k > X^\kappa$, the cofactor mass bound and $m-n \neq 0$ yield a Ramanujan tail of size $X^{-\kappa+o(1)}$. Consequently, for every fixed $A > 0$ and every fixed $0 < \kappa < \min\{s, \frac{1}{2}(\frac{1}{2} - \delta)\}$,

$$K_S^0(m, n) \ll_A (\log X)^{-A} + L^{-2}(\log \log(3X))^3 + X^{-\kappa+o(1)}.$$

The actual physical row coefficients have $O(Q)$ one-row absolute mass. Extracting frequency zero before Mellin separation therefore closes the physical base zero mode. Uniform endpoint calibration also closes the three scalar channels EZ, ZE, EE . Combining these facts with the inherited static or dynamic exact-conductor theorem proves

$$\mathcal{Q}_S^{\text{phys}} \ll_B XQ(\log X)^{-B}$$

throughout its strict Poisson-short base region. Splicing this base estimate with the annular cutoff-stability theorem closes the selected calibrated square at certain $T > J$. In particular, the rational sample

$$\delta = \frac{3}{20}, \quad \beta = \frac{31}{50}, \quad s = \frac{7}{20}, \quad t = \frac{39}{100}$$

is upgraded from conditional cutoff stability to a closed selected truncated square. The higher- β sample whose base already lies beyond the Poisson length is not covered. No ultra-long complement, positivity statement, Hardy–Littlewood asymptotic, twin-prime theorem, or general parity breakthrough is claimed.

Keywords: primitive Möbius tail; calibrated base; additive zero mode; Ramanujan sum; Poisson boundary; sieve parity.

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1 Introduction

The primitive Möbius-tail program of TPC-20–26 rewrites a selected fixed-shift prime-pair residual as a bilinear correlation of opened rows

$$m = \ell d, \quad \ell \asymp L \text{ prime}, \quad d \asymp D, \quad Q = LD, \quad QJ \asymp X.$$

The common residual divisor coefficient is

$$a(u) = -\mu(u) \log u, \quad b_R(u) = a(u) - \lambda'_R(u), \quad \lambda'_R(u) = 0 \quad (u > R). \quad (1)$$

TPC-20 derives the exact four-channel centered kernel, TPC-21 closes its multiplicative-character mean at nonzero additive frequency, and TPC-22–23 close the corresponding connected fiber discrepancy in explicit short-support regions [2–5]. TPC-24–26 then use endpoint calibration and exact-conductor estimates to pass selected cutoffs across the old Poisson face [6–8].

There is a precise logical gap between those statements. The scalar packets displayed and estimated in TPC-22–23 are sums over $r \neq 0$. Their formulas do not estimate the additive zero term retained by the TPC-20 two-scale completion. Moreover, the physical prefix used in TPC-26 is calibrated rather than merely centered. Thus one cannot infer a physical base estimate by replacing the words “centered selected square” with “calibrated square.” Two separate operations are required:

- (i) close the connected additive zero mode of the centered square;
- (ii) restore the scalar calibration channels without losing the natural scale.

The present paper performs both operations for the actual physical opened-row provenance.

1.1 The exact zero-mode factorization

For one row put

$$Z_{m,S}(j) = \sum_{\substack{u \leq S \\ (u, mH)=1}} b_R(u) C_{m,u}(j), \quad C_{m,u}(j) = \mathbf{1}_{u|mj+h} - \frac{1}{u}. \quad (2)$$

If $m \neq n$, the primitive complete-period average of $C_{m,u}C_{n,v}$ is

$$\frac{1}{uv} \left(g \mathbf{1}_{g|m-n} - 1 \right), \quad g = (u, v). \quad (3)$$

The elementary Ramanujan identity

$$g \mathbf{1}_{g|\Delta} - 1 = \sum_{\substack{k|g \\ k>1}} c_k(\Delta) \quad (4)$$

turns the two-divisor zero kernel into a sum of products of one-row multiple densities:

$$\boxed{K_S^0(m, n) = \sum_{k>1} c_k(m-n) B_{k,S}(m) B_{k,S}(n).} \quad (5)$$

Here

$$B_{k,S}(m) = \sum_{\substack{u \leq S \\ k|u \\ (u, mH)=1}} \frac{b_R(u)}{u}. \quad (6)$$

The row coprimality restrictions separate exactly; there is no approximation in (5).

This factorization changes the nature of the problem. It is not necessary to estimate the centered two-divisor kernel by an operator norm. Instead, one calibrates each short one-row multiple density. For squarefree k coprime to the row, the finite model satisfies an exact identity

$$\sum_{\substack{u \leq R \\ k|u \\ (u, mH)=1}} \frac{\lambda'_R(u)}{u} = \frac{\mu(k)}{\varphi(k)} \rho_{M,H,R}(m) \quad (7)$$

whenever $kdH \leq R$. The corresponding prime coefficient has main term

$$\sum_{\substack{u \leq S \\ k|u \\ (u, mH)=1}} \frac{a(u)}{u} = \frac{\mu(k)}{\varphi(k)} \rho_{P,H}(m) + O_A \left(\frac{\rho_{P,H}(m)(1 + \log k)}{\varphi(k)(\log X)^A} \right). \quad (8)$$

Their difference is the already power-saving row drift

$$\delta_H(m) = \rho_{P,H}(m) - \rho_{M,H,R}(m) \ll \frac{\log \log(3X)}{L}.$$

Choose $K_0 = X^\kappa$ below $R^{1/2}$. The identities above control $k \leq K_0$. For $k > K_0$, the cofactor estimate

$$|B_{k,S}(m)| \ll \frac{(\log X)^4}{k} \quad (9)$$

and the off-diagonal relation $m - n \neq 0$ give

$$\sum_{k>K_0} \frac{|c_k(m-n)|}{k^2} \ll X^{o(1)} K_0^{-1}. \quad (10)$$

This proves the pointwise zero theorem

$$K_S^0(m, n) \ll_A (\log X)^{-A} + L^{-2} (\log \log(3X))^3 + X^{-\kappa+o(1)}. \quad (11)$$

The condition $m \neq n$ is structural. When $m = n$, one has $c_k(0) = \varphi(k)$, and the large- k argument does not give (10).

1.2 From the centered base to the physical prefix

TPC-24's exact calibration identity on the primitive orbit $(j, H) = 1$ is

$$P_{m,S}(j) = Z_{m,S}(j) + E_S(m), \quad E_S(m) = A_{mH}(S) - \rho_{P,H}(m). \quad (12)$$

The classical zero-free region gives $E_S(m) \ll_A (\log X)^{-A}$, uniformly in the polynomial row ranges used here. Therefore

$$P_{m,S}P_{n,S} = ZZ + EZ + ZE + EE. \quad (13)$$

The last three terms are bounded before spectral separation. The actual one-row absolute masses are $O(Q)$, while

$$\sum_j |Z_{m,S}(j)| |F(j/J)| \ll (J+S)(\log(2X))^4.$$

Thus the scalar channels have arbitrary logarithmic saving whenever $S \leq JX^{-\eta_0}$. This is precisely the strict Poisson-short condition already required by the dynamic nonzero theorem.

For the ZZ term, physical frequency zero is extracted before Mellin separation. The new pointwise bound (11) combines with the actual $O(Q^2)$ row-pair absolute mass. Only the nonzero spectrum is then sent through the applicable TPC-22 static or TPC-23 dynamic conductor theorem. The correct order is

extract and factor the centered physical zero mode $\longrightarrow$ calibrate its one-row multiples $\longrightarrow$ separate the nonzero spectrum $\longrightarrow$ restore EZ, ZE, EE .
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(14)

1.3 Consequences and boundary

Write

$$Q = X^{\beta+o(1)}, \quad J = X^{1-\beta+o(1)}, \quad S = X^{s+o(1)}. \quad (15)$$

Under the inherited no-wrap, physical provenance, off-diagonal mask, and fixed-margin hypotheses, the base closes when

$$s < 1 - \beta, \quad \eta_S(\beta, s) = \min \left\{ 1 - \frac{3\beta}{2}, \frac{\beta + 1 - 4s}{2}, \frac{3 - \beta - 6s}{4} \right\} > 0. \quad (16)$$

Combining this with the TPC-26 annular theorem closes a selected calibrated square at an upper cutoff $T = X^{t+o(1)} > J$ whenever its annular minimax $M_\beta(t)$ is positive. The sample

$$(\delta, \beta, s, t) = \left(\frac{3}{20}, \frac{31}{50}, \frac{7}{20}, \frac{39}{100} \right) \quad (17)$$

has centered nonzero conductor ledger margin $7/100$ and annular minimax margin $1/100$; the complete calibrated conclusion has arbitrary logarithmic saving.

This does not make the higher- β cutoff-stability sample of TPC-26 into a closed square. In that sample $S > J$, so the inherited base nonzero theorem is unavailable. Nor does the argument estimate the ultra-long complement after the upper cutoff. It proves one selected physical upper-bound/cancellation statement; it does not produce positivity, a Hardy–Littlewood main term, twin primes, or a general breach of sieve parity.

The paper is organized as follows. [Section 2](#) fixes the physical packet and the exact centered-to-calibrated bridge. [Section 3](#) proves the zero-mode factorization. [Section 4](#) proves the exact model-multiple identity and its calibrated prime companion. [Section 5](#) obtains the pointwise and physical zero theorems. [Section 6](#) transfers the inherited nonzero theorem and closes the scalar channels. [Section 7](#) gives the base and upper-cutoff closure, and [section 8](#) records the exact certificate and strict scope.

2 Physical interface and the exact calibration bridge

We retain the actual opened-row provenance before any Mellin or Fourier separation. This is essential only at physical additive frequency zero; the nonzero spectrum will be separated later.

2.1 Rows, scales, and coefficients

Fix $h \in \mathbb{Z} \setminus \{0\}$, and put

$$H = \text{rad } |h|, \quad H_0 = |h|. \quad (18)$$

Let

$$R = X^{r_0+o(1)}, \quad r_0 = \frac{1}{2} - \delta > 0, \quad R \leq S \leq X^{C_S}, \quad (19)$$

where all exponent inequalities below have fixed positive margins. In one opened slice, rows are indexed by

$$\alpha = (\ell, d_\alpha), \quad m_\alpha = \ell d_\alpha \asymp Q = LD, \quad (20)$$

with

$$\ell \in [L/2, 2L] \text{ prime}, \quad d_\alpha \in [D, 2D] \cap [1, R^{1/2}], \quad d_\alpha \text{ squarefree}, \quad (d_\alpha, H) = 1. \quad (21)$$

We assume

$$QJ \asymp X, \quad L > \max\{2R, 2H_0\}, \quad L \geq X^{\lambda_0} \quad (22)$$

for a fixed $\lambda_0 > 0$. In particular, $\ell > R$. On the primitive orbit $(j, H) = 1$, target primitivity gives

$$(u, m_\alpha H) = 1 \quad \text{whenever } u \mid m_\alpha j + h \text{ and } (j, H) = 1. \quad (23)$$

Since H is fixed and $d_\alpha \leq R^{1/2}$, we also have $d_\alpha H \leq R$ for all sufficiently large X . This is the finite-model completeness condition used below.

The two physical row coefficients have the form

$$\gamma_\alpha^{(i)} = \mu(d_\alpha)(\log \ell) \omega_D^{(i)}(d_\alpha) \psi_L^{(i)}(\ell/L) \zeta_\alpha^{(i)}, \quad |\zeta_\alpha^{(i)}| \ll 1, \quad i = 1, 2, \quad (24)$$

where the smooth and fixed-residue data range over fixed bounded families. Chebyshev's estimate gives

$$\sum_\alpha |\gamma_\alpha^{(i)}| \ll Q, \quad (25)$$

$$\|\gamma^{(i)}\|_\infty \ll \log(2L), \quad \|\gamma^{(i)}\|_2^2 \ll Q \log(2L). \quad (26)$$

These are the provenance bounds of TPC-25 [7]. The generic row-pair mask satisfies

$$|\mathbf{m}(\alpha, \gamma)| \leq 1, \quad \mathbf{m}(\alpha, \gamma) = 0 \quad \text{if } m_\alpha = m_\gamma, \quad (27)$$

and is independent of all divisor and conductor variables. The second condition will be used in the large-Ramanujan-modulus tail.

Let $\Xi_{\alpha, \gamma}(j)$ be a uniformly bounded real periodic factor of fixed period H_0 . The row-orbit weight $\mathcal{W}_{\alpha, \gamma}(j/J)$ belongs to the factorable physical family of TPC-25. It is a uniformly bounded C_c^∞ family supported in one fixed compact interval, so its sampled orbit has length $O(J)$. In particular,

$$\sup_{\alpha, \gamma} \left| \int_{\mathbb{R}} \mathcal{W}_{\alpha, \gamma}(t) dt \right| \ll 1, \quad (28)$$

and continuous Mellin separation has finite weighted variation in every fixed orbit seminorm. We denote all fixed shift, smoothness, residue, source, and exponent-margin data by $\mathcal{D}$.

2.2 Finite model, prime density, and row drift

Put

$$a(u) = -\mu(u) \log u \quad (29)$$

and

$$\lambda'_R(u) = u \sum_{b \leq R/u} \frac{\mu(b) \mu(u)}{\varphi(b)} \quad (u \leq R), \quad \lambda'_R(u) = 0 \quad (u > R). \quad (30)$$

Define

$$b_R(u) = a(u) - \lambda'_R(u). \quad (31)$$

All three coefficients vanish off squarefree integers. The finite-model and prime conditional densities are

$$\rho_{M,H,R}(m_\alpha) = \frac{H}{\varphi(H)} \frac{d_\alpha}{\varphi(d_\alpha)}, \quad (32)$$

$$\rho_{P,H}(m_\alpha) = \frac{m_\alpha H}{\varphi(m_\alpha H)}. \quad (33)$$

Since $\ell > R$, the large row prime occurs only in the second density. Their difference is

$$\delta_H(m_\alpha) := \rho_{P,H}(m_\alpha) - \rho_{M,H,R}(m_\alpha) = \frac{H}{\varphi(H)} \frac{d_\alpha}{\varphi(d_\alpha)(\ell-1)} \ll_h \frac{\log \log(3X)}{L}. \quad (34)$$

For $M, Y \geq 1$, write

$$A_M(Y) = \sum_{\substack{u \leq Y \\ (u, M)=1}} \frac{-\mu(u) \log u}{u}, \quad (35)$$

$$C_M(Y) = \sum_{\substack{u \leq Y \\ (u, M)=1}} \frac{\mu(u)}{u}. \quad (36)$$

The classical zero-free region for $\zeta(s)$, in the uniform form proved and used in TPC-20 and TPC-24, gives for every fixed $A, C > 0$

$$A_M(Y) = \frac{M}{\varphi(M)} + O_{A,C} \left(\frac{M}{\varphi(M)} (\log Y)^{-A} \right), \quad (37)$$

$$C_M(Y) \ll_{A,C} \frac{M}{\varphi(M)} (\log Y)^{-A}, \quad (38)$$

uniformly for $Y \geq 3$ and $M \leq Y^C$ [1, 2, 6]. These are proved theorems, not conjectural distribution hypotheses.

2.3 Centered and calibrated prefixes

For $R \leq S$, define

$$Z_{m,S}(j) := \sum_{\substack{u \leq S \\ (u, mH)=1}} b_R(u) \left(\mathbf{1}_{u|mj+h} - \frac{1}{u} \right), \quad (39)$$

$$E_S(m) := A_{mH}(S) - \rho_{P,H}(m), \quad (40)$$

$$P_{m,S}(j) := \sum_{\substack{u \leq S \\ u|mj+h}} b_R(u) - \delta_H(m). \quad (41)$$

Proposition 2.1 (Exact centered-to-calibrated bridge). *For every retained row, every $R \leq S$, and every primitive orbit point $(j, H) = 1$,*

$$\boxed{P_{m,S}(j) = Z_{m,S}(j) + E_S(m)}. \quad (42)$$

Moreover, for every fixed $A > 0$,

$$\boxed{E_S(m) \ll_{A,\mathcal{O}} (\log X)^{-A}} \quad (43)$$

uniformly in the retained polynomial row and cutoff ranges.

Proof. The finite model has the exact density identity

$$\sum_{\substack{u \leq R \\ (u, mH)=1}} \frac{\lambda'_R(u)}{u} = \rho_{M,H,R}(m). \quad (44)$$

Since λ'_R is supported on $u \leq R$, subtraction from $A_{mH}(S)$ gives

$$\sum_{\substack{u \leq S \\ (u, mH)=1}} \frac{b_R(u)}{u} = \delta_H(m) + E_S(m). \quad (45)$$

Expand the centered indicator in (39), insert (45), and use target primitivity (23). This proves (42). Estimate (43) is (37) with $M = mH$, since all relevant quantities are fixed powers of X . The factor $mH/\varphi(mH) \ll_h \log \log(3X)$ is absorbed by first invoking (37) with one additional logarithmic power. $\square$

The physical base packet is

$$\begin{aligned} \mathcal{Q}_S^{\text{phys}} &:= \sum_{\alpha, \gamma} \gamma_{\alpha}^{(1)} \gamma_{\gamma}^{(2)} \mathbf{m}(\alpha, \gamma) \\ &\quad \times \sum_{\substack{j \in \mathbb{Z} \\ (j, H)=1}} \Xi_{\alpha, \gamma}(j) \mathcal{W}_{\alpha, \gamma}(j/J) P_{m_{\alpha}, S}(j) P_{m_{\gamma}, S}(j). \end{aligned} \quad (46)$$

The natural scale is

$$JQ^2 \asymp XQ. \quad (47)$$

By proposition 2.1, the integrand in (46) has the exact four-term decomposition

$$P_{m,S} P_{n,S} = Z_{m,S} Z_{n,S} + E_S(m) Z_{n,S} + Z_{m,S} E_S(n) + E_S(m) E_S(n). \quad (48)$$

The first term contains the new additive-zero problem. The last three will be closed directly in section 6.

3 Ramanujan-square factorization of the connected zero mode

The base zero mode is finite and admits an exact one-row factorization. This section contains no asymptotic estimate.

3.1 Primitive complete-period covariance

For admissible rows and divisors, recall

$$C_{m,u}(j) = \mathbf{1}_{u|mj+h} - \frac{1}{u}.$$

Let $\mathbb{E}_H$ denote the complete-period average conditional on $(j, H) = 1$. The primitive CRT calculation of TPC-20 gives

$$\mathbb{E}_H[C_{m,u}C_{n,v}] = \frac{1}{uv} \left(g \mathbf{1}_{g|m-n} - 1 \right), \quad g = (u, v), \quad (49)$$

provided $(u, mH) = (v, nH) = 1$ [2]. The normalization in (49) omits the fixed outer density $\varphi(H)/H$; that density is restored in the physical Poisson zero coefficient and is harmless.

Define the centered base zero kernel

$$K_S^0(m, n) := \sum_{\substack{u, v \leq S \\ (u, mH)=1 \\ (v, nH)=1}} \frac{b_R(u)b_R(v)}{uv} \left((u, v) \mathbf{1}_{(u,v)|m-n} - 1 \right). \quad (50)$$

It is exactly the arithmetic factor multiplying the physical additive zero functional for $Z_{m,S}Z_{n,S}$.

Remark 3.1 (Fixed residue cells). The physical factor $\Xi_{\alpha, \gamma}$ may first be split into its finitely many classes $j \equiv \varrho \pmod{H_0}$. Every effective divisor is coprime to H_0 , so the joint and one-leg densities inside one such class are respectively $1/(H_0[u, v])$, $1/(H_0u)$, and $1/(H_0v)$. Thus (50) is unchanged up to a bounded fixed local factor. No growing residue complexity is hidden here.

3.2 The divisor identity

We use the Ramanujan sum

$$c_k(a) = \sum_{\substack{x \bmod k \\ (x, k)=1}} e_k(ax). \quad (51)$$

Lemma 3.2 (Ramanujan resolution of compatibility). *For all positive integers g and all $\Delta \in \mathbb{Z}$,*

$$\boxed{g \mathbf{1}_{g|\Delta} - 1 = \sum_{\substack{k|g \\ k>1}} c_k(\Delta)}. \quad (52)$$

Proof. The standard divisor formula

$$c_k(\Delta) = \sum_{e|(k, \Delta)} e \mu(k/e) \quad (53)$$

gives

$$\sum_{k|g} c_k(\Delta) = \sum_{e|(g, \Delta)} e \sum_{r|g/e} \mu(r).$$

The inner sum is nonzero only when $e = g$, in which case it equals one. Hence the full sum is g if $g | \Delta$ and zero otherwise. Subtract the $k = 1$ term $c_1(\Delta) = 1$. $\square$

For $k \geq 1$, define the one-row multiple density

$$B_{k,S}(m) := \sum_{\substack{u \leq S \\ k|u \\ (u, mH)=1}} \frac{b_R(u)}{u}. \quad (54)$$

Because b_R is squarefree supported, $B_{k,S}(m) = 0$ unless k is squarefree and $(k, mH) = 1$.

Theorem 3.3 (Exact Ramanujan-square factorization). *For every pair of rows m, n ,*

$$\boxed{K_S^0(m, n) = \sum_{k>1} c_k(m - n) B_{k,S}(m) B_{k,S}(n).} \quad (55)$$

The sum is finite: only squarefree $k \leq S$ coprime to mnH can contribute.

Proof. Insert lemma 3.2 with $g = (u, v)$ and $\Delta = m - n$ into (50). All sums are finite, so their order may be interchanged. The condition $k \mid (u, v)$ is the product of the two independent conditions $k \mid u$ and $k \mid v$. The row coprimality restrictions also separate, giving

$$K_S^0(m, n) = \sum_{k>1} c_k(m - n) \left(\sum_{\substack{u \leq S \\ k \mid u \\ (u, mH)=1}} \frac{b_R(u)}{u} \right) \left(\sum_{\substack{v \leq S \\ k \mid v \\ (v, nH)=1}} \frac{b_R(v)}{v} \right).$$

This is (55). □

Remark 3.4 (What the factorization removes). The coherent $g = 1$ zero block vanishes identically in the centered base: its multiplier is $1 - 1 = 0$. This is different from an uncentered one-sided shell kernel, where a nonzero $g = 1$ principal row direction may remain. The sum in (55) therefore starts at $k > 1$ for an exact algebraic reason, not because a principal mode has been deleted.

3.3 Two Ramanujan ledgers

The following elementary bounds will be used after calibration.

Lemma 3.5 (Ramanujan weight and tail). *Let $0 < |\Delta| \leq X^C$, and let $2 \leq K \leq X^C$. Then*

$$\sum_{\substack{k \geq 1 \\ k \text{ squarefree}}} \frac{|c_k(\Delta)|}{\varphi(k)^2} \ll \frac{\text{rad } |\Delta|}{\varphi(\text{rad } |\Delta|)} \ll_C \log \log(3|\Delta|), \quad (56)$$

$$\sum_{k>K} \frac{|c_k(\Delta)|}{k^2} \ll \frac{\tau(|\Delta|)}{K}. \quad (57)$$

The second estimate remains true after restricting to squarefree k .

Proof. For squarefree k , multiplicativity and the prime identity

$$|c_p(\Delta)| = \begin{cases} p - 1, & p \mid \Delta, \\ 1, & p \nmid \Delta \end{cases}$$

give the absolutely convergent Euler product

$$\begin{aligned} \sum_{\substack{k \geq 1 \\ k \text{ squarefree}}} \frac{|c_k(\Delta)|}{\varphi(k)^2} &= \prod_{p \mid \Delta} \left(1 + \frac{1}{p-1} \right) \prod_{p \nmid \Delta} \left(1 + \frac{1}{(p-1)^2} \right) \\ &\ll \prod_{p \mid \Delta} \frac{p}{p-1} = \frac{\text{rad } |\Delta|}{\varphi(\text{rad } |\Delta|)}. \end{aligned}$$

The standard elementary maximal-order bound for $n/\varphi(n)$ proves the last estimate in (56).

For the tail, formula (53) implies

$$|c_k(\Delta)| \leq (k, \Delta) \leq \sum_{e|(k, \Delta)} \varphi(e). \quad (58)$$

Apply this bound and write $k = er$:

$$\sum_{k>K} \frac{|c_k(\Delta)|}{k^2} \leq \sum_{e|\Delta} \frac{\varphi(e)}{e^2} \sum_{r>K/e} \frac{1}{r^2}.$$

If $e \leq K$, the inner sum is $O(e/K)$; if $e > K$, it is $O(1)$. Both ranges contribute $O(\tau(|\Delta|)/K)$, proving (57). $\square$

The second estimate fails in the form required here when $\Delta = 0$, because $c_k(0) = \varphi(k)$. This is precisely why the physical off-diagonal mask in (27) cannot be removed before the zero-mode estimate.

4 Exact multiple densities and uniform cofactor calibration

We now estimate the factors $B_{k,S}(m)$ in (55). The main cancellation is between an exact finite-model identity and the main term of a proved uniform Möbius calibration theorem.

4.1 An exact finite-model multiple identity

Let $m = \ell d_m$ be one retained row and put

$$M_m = d_m H. \quad (59)$$

The integer M_m is squarefree, while $\ell > R$.

Proposition 4.1 (Exact model density on multiples). *Let k be squarefree and $(k, mH) = 1$. Then*

$$\sum_{\substack{u \leq R \\ k|u \\ (u, mH)=1}} \frac{\lambda'_R(u)}{u} = \frac{\mu(k)}{\varphi(k)} \sum_{\substack{a|M_m \\ ak \leq R}} \frac{1}{\varphi(a)}. \quad (60)$$

In particular, if $kM_m \leq R$, then

$$\sum_{\substack{u \leq R \\ k|u \\ (u, mH)=1}} \frac{\lambda'_R(u)}{u} = \frac{\mu(k)}{\varphi(k)} \rho_{M,H,R}(m). \quad (61)$$

Proof. From (30), the left side of (60) is

$$\sum_{\substack{ub \leq R \\ k|u \\ (u, mH)=1}} \frac{\mu(ub)\mu(b)}{\varphi(ub)}. \quad (62)$$

Only squarefree $q = ub$ contributes. Grouping by q gives

$$\sum_{q \leq R} \frac{\mu(q)}{\varphi(q)} \sum_{\substack{b|q \\ k|q/b \\ (q/b, mH)=1}} \mu(b). \quad (63)$$

Because $\ell > R$, no $q \leq R$ contains the large row prime. Put $a = (q, M_m)$. The condition $(q/b, M_m) = 1$ forces $a \mid b$. Since $(k, M_m) = 1$, write $q = aq_0$ and $b = ab_0$. The inner sum becomes

$$\mu(a) \sum_{b_0 \mid q_0/k} \mu(b_0), \quad (64)$$

and is zero unless $q_0 = k$, in which case it equals $\mu(a)$. Thus $q = ak$ with $a \mid M_m$, and the surviving summand is

$$\frac{\mu(ak)\mu(a)}{\varphi(ak)} = \frac{\mu(k)}{\varphi(k)\varphi(a)}. \quad (65)$$

This proves (60). If $kM_m \leq R$, every divisor $a \mid M_m$ occurs, and

$$\sum_{a \mid M_m} \frac{1}{\varphi(a)} = \prod_{p \mid M_m} \left(1 + \frac{1}{p-1}\right) = \frac{M_m}{\varphi(M_m)} = \rho_{M,H,R}(m). \quad (66)$$

This proves (61). $\square$

The cutoff condition in this proposition is why the opened-row bound $d_m \leq R^{1/2}$ matters. It leaves a fixed-power range of multipliers k on which the model identity is complete.

4.2 The matching prime multiple

Define

$$A_{k,S}^{\text{mult}}(m) := \sum_{\substack{u \leq S \\ k \mid u \\ (u, mH)=1}} \frac{a(u)}{u}. \quad (67)$$

Lemma 4.2 (Prime multiple calibration). *Let k be squarefree and $(k, mH) = 1$. Then the exact identity*

$$A_{k,S}^{\text{mult}}(m) = \frac{\mu(k)}{k} \{A_{kmH}(S/k) - (\log k)C_{kmH}(S/k)\} \quad (68)$$

holds. Fix $0 < \kappa < s$, where $S = X^{s+o(1)}$. Uniformly for $k \leq X^\kappa$ and every fixed $A > 0$,

$$A_{k,S}^{\text{mult}}(m) = \frac{\mu(k)}{\varphi(k)} \rho_{P,H}(m) + O_{A,\mathcal{D}} \left(\frac{\rho_{P,H}(m)(1 + \log k)}{\varphi(k)(\log X)^A} \right). \quad (69)$$

Proof. On the squarefree support, write $u = kw$. Then $(w, kmH) = 1$ and

$$a(kw) = \mu(k)a(w) - \mu(k)\mu(w) \log k. \quad (70)$$

Division by kw and summation prove (68).

For $k \leq X^\kappa$, the calibration length $Y = S/k$ is at least a fixed positive power of X . Also $kmH \leq X^{O(1)}$, so one fixed constant C , depending only on the exponent data, gives $kmH \leq Y^C$. Apply equations (37) and (38) with a calibration exponent larger than the requested A . Since $(k, mH) = 1$, the main term is

$$\frac{\mu(k)}{k} \frac{kmH}{\varphi(kmH)} = \frac{\mu(k)}{\varphi(k)} \rho_{P,H}(m). \quad (71)$$

The two error terms are bounded by the displayed remainder because $\log(S/k) \asymp \log X$ uniformly in this fixed-power range. This proves (69). $\square$

4.3 Small and large multiple densities

Set

$$K_0 = X^{\kappa+o(1)}, \quad 0 < \kappa < \min \left\{ s, \frac{r_0}{2} \right\}. \quad (72)$$

Because $d_m \leq R^{1/2}$ and H is fixed, all sufficiently large X satisfy

$$kM_m \leq R \quad (k \leq K_0). \quad (73)$$

Corollary 4.3 (Calibrated small multiples). *For every retained row, every effective squarefree $k \leq K_0$, and every fixed $A > 0$,*

$$\boxed{B_{k,S}(m) = \frac{\mu(k)}{\varphi(k)} \delta_H(m) + O_{A,\mathscr{D}} \left(\frac{1}{\varphi(k)(\log X)^A} \right)}. \quad (74)$$

Proof. By (31), $B_{k,S}(m)$ is the prime multiple (67) minus the model multiple. Apply [proposition 4.1](#) and [lemma 4.2](#) and use (34). Since $\rho_{P,H}(m) \ll_h \log \log(3X)$ and $1 + \log k \ll \log X$, invoke [lemma 4.2](#) with three additional logarithmic powers and then rename the exponent. This absorbs the explicit polylogarithmic factors without introducing an X^ε loss. $\square$

For large k , cancellation is unnecessary. We import the elementary cofactor mass lemma of TPC-24: uniformly for squarefree $k \leq S$,

$$\sum_{\substack{w \leq S/k \\ (w,k)=1}} \frac{|b_R(kw)|}{w} \ll_{C_S} (\log(2X))^4. \quad (75)$$

It follows from the positive divisor formula for λ'_R and harmonic summation, and uses no cancellation [6].

Lemma 4.4 (Absolute large-multiple bound). *Uniformly for every row and every $k \geq 1$,*

$$\boxed{|B_{k,S}(m)| \ll_{\mathscr{D}} \frac{(\log(2X))^4}{k}}. \quad (76)$$

Proof. There is nothing to prove unless k is squarefree and $(k, mH) = 1$. Write $u = kw$ in (54). Squarefree support forces $(w, k) = 1$, while the remaining row-coprimality condition only decreases the absolute sum. Thus

$$|B_{k,S}(m)| \leq \frac{1}{k} \sum_{\substack{w \leq S/k \\ (w,k)=1}} \frac{|b_R(kw)|}{w},$$

and (75) proves the result. $\square$

The two bounds (74) and (76) concern the same exact one-row factor. The split at K_0 therefore loses no intermediate term.

5 Pointwise and physical closure of the base zero mode

We combine the exact factorization with the two multiple-density estimates. The only use of the row off-diagonal condition is in the large- k Ramanujan tail.

5.1 The pointwise theorem

Theorem 5.1 (Pointwise centered base-zero theorem). *Retain the row and cutoff hypotheses of [section 2](#), and let $K_0 = X^{\kappa+o(1)}$ satisfy [\(72\)](#). Uniformly for distinct retained rows $m \neq n$ and every fixed $A > 0$,*

$$\boxed{K_S^0(m, n) \ll_{A, \mathcal{D}} (\log X)^{-A} + L^{-2}(\log \log(3X))^3 + X^{-\kappa+o(1)}}. \quad (77)$$

No mean-zero condition on the row weights or the physical orbit test is assumed.

Proof. Use the exact factorization [\(55\)](#) and split its sum at K_0 . For $k \leq K_0$, write

$$B_{k,S}(m) = \frac{\mu(k)}{\varphi(k)} \delta_H(m) + e_k(m), \quad |e_k(m)| \ll_A \frac{1}{\varphi(k)(\log X)^{A+10}}. \quad (78)$$

The harmless extra ten logarithms anticipate the fixed reassembly losses. By [\(34\)](#),

$$|B_{k,S}(m)B_{k,S}(n)| \ll_A \frac{1}{\varphi(k)^2} \left\{ \frac{(\log \log(3X))^2}{L^2} + (\log X)^{-A-5} \right\}. \quad (79)$$

Apply [\(56\)](#) and increase the calibration exponent once more. The small- k contribution is

$$\ll_A L^{-2}(\log \log(3X))^3 + (\log X)^{-A}. \quad (80)$$

For $k > K_0$, [lemma 4.4](#) gives

$$|B_{k,S}(m)B_{k,S}(n)| \ll \frac{(\log(2X))^8}{k^2}. \quad (81)$$

Since $m \neq n$ and $|m - n| \leq X^{O(1)}$, apply [\(57\)](#) with $\Delta = m - n$. The large- k contribution is

$$\ll X^{-\kappa+o(1)}. \quad (82)$$

Combining [\(80\)](#) and [\(82\)](#) proves the theorem. $\square$

Remark 5.2 (Why no operator hypothesis remains). Entrywise decay alone does not close a dense bilinear row matrix under only soft ℓ^2 energy; TPC-25 gives an exact interface counterexample. The present physical packet has more information: its one-row ℓ^1 -mass is $O(Q)$. We use that actual provenance before separation. Thus [theorem 5.1](#) does not contradict the earlier operator obstruction and does not assert a black-box theorem for arbitrary soft-energy row families.

5.2 Physical frequency zero before separation

Let $\mathcal{Z}_S^{ZZ,0}$ denote the physical additive-zero contribution of the $Z_{m_\alpha,S}Z_{m_\gamma,S}$ term in [\(48\)](#). It is taken before continuous Mellin separation. After splitting the $O_h(1)$ residue cells described in [remark 3.1](#), it has the form

$$\mathcal{Z}_S^{ZZ,0} = J \sum_{\alpha, \gamma} \gamma_\alpha^{(1)} \gamma_\gamma^{(2)} \mathbf{m}(\alpha, \gamma) \vartheta_{\alpha, \gamma}^{(0)} K_S^0(m_\alpha, m_\gamma), \quad (83)$$

where

$$\vartheta_{\alpha, \gamma}^{(0)} = \left(\frac{1}{H_0} \sum_{\substack{\varrho \bmod H_0 \\ (\varrho, H)=1}} \Xi_{\alpha, \gamma}(\varrho) \right) \int_{\mathbb{R}} \mathcal{W}_{\alpha, \gamma}(t) dt, \quad |\vartheta_{\alpha, \gamma}^{(0)}| \ll_{\mathcal{D}} 1. \quad (84)$$

Thus the fixed primitive density, residue-cell coefficient, and zero Fourier functional occur exactly once. The bound in [\(84\)](#) follows from [\(28\)](#).

Theorem 5.3 (Physical centered base-zero closure). *Under the hypotheses of [theorem 5.1](#), for every fixed $B > 0$,*

$$\boxed{\mathcal{Z}_S^{ZZ,0} \ll_{B,\mathscr{D}} XQ(\log X)^{-B}}. \quad (85)$$

Proof. The mask in (27) restricts (83) to $m_\alpha \neq m_\gamma$, so [theorem 5.1](#) applies. Delete the bounded mask and zero functional, then use (25) on both row sides. This gives

$$|\mathcal{Z}_S^{ZZ,0}| \ll_{A,\mathscr{D}} JQ^2 \left\{ (\log X)^{-A} + L^{-2}(\log \log(3X))^3 + X^{-\kappa+o(1)} \right\}. \quad (86)$$

By (47), $JQ^2 \asymp XQ$. The last two terms have fixed-power savings because $L \geq X^{\lambda_0}$ and $\kappa > 0$. Choose A after B and after the fixed logarithmic residue/dyadic multiplicities. This proves (85). $\square$

Remark 5.4 (No mean-zero physical test). The proof uses only the bounded zero functional (28); it does not require it to vanish. Thus the physical zero mode is closed for a general fixed nonzero-mean test in the factorable provenance class.

Remark 5.5 (Diagonal boundary). If the row diagonal were restored, the large- k argument would face $c_k(0) = \varphi(k)$, and (57) would no longer supply K_0^{-1} . The theorem therefore closes the actual generic off-diagonal packet. It is not a diagonal second-moment theorem.

6 Nonzero transfer and scalar reassembly

The new work at additive frequency zero is complete. We now transfer the inherited exact-conductor theorem to the same physical packet and then restore the three calibration channels in (48).

6.1 The inherited dynamic conductor region

Write

$$Q = X^{\beta+o(1)}, \quad J = X^{1-\beta+o(1)}, \quad S = X^{s+o(1)}. \quad (87)$$

For the selected square $u, v \leq S$, the common exact-conductor ledger of TPC-22 at $S = R$ and TPC-23 at $S > R$ is

$$\eta_S(\beta, s) = \min \left\{ 1 - \frac{3\beta}{2}, \frac{\beta + 1 - 4s}{2}, \frac{3 - \beta - 6s}{4} \right\}. \quad (88)$$

The corresponding nonzero theorem assumes

$$s < 1 - \beta, \quad \eta_S(\beta, s) > 0 \quad (89)$$

with fixed margins, together with the primitive row interface, divisor-independent off-diagonal mask, no-wrap condition, and the stated smooth Fourier-tail control [5]. The first inequality is the strict Poisson-short condition

$$HS \leq JX^{-\eta_0} \quad (90)$$

for some fixed $\eta_0 > 0$, after adjusting fixed constants.

At the boundary $S = R$ we use the static theorem of TPC-22; in the genuinely dynamic case $S > R$ we use TPC-23. The theorem acts on the retained nonzero joint packet. Its global principal multiplicative-character cell is bounded absolutely by $Q^2 X^\varepsilon$, while every nonprincipal exact-conductor cell has a fixed-power saving from (88). It does not include the additive zero kernel; that is why [theorem 5.3](#) was needed.

6.2 Transfer of the nonzero spectrum

Let $\mathcal{N}_S^{ZZ, \neq 0}$ be the complete physical contribution of all nonzero additive frequencies from the centered product $Z_{m_\alpha, S} Z_{m_\gamma, S}$, after the additive zero term (83) has been set aside.

Lemma 6.1 (Physical transfer to the dynamic interface). *After splitting the $O_h(1)$ residue cells, the physical nonzero packet has a continuous representation*

$$\gamma_\alpha^{(1)} \gamma_\gamma^{(2)} \mathcal{W}_{\alpha, \gamma}(j/J) = \int x_\tau(\alpha) y_\tau(\gamma) \Phi_\tau(j/J) d\nu(\tau), \quad (91)$$

such that, for every fixed integer $M \geq 0$,

$$\int (1 + |\tau|)^M \|x_\tau\|_1 \|y_\tau\|_1 d|\nu|(\tau) \ll_{M, \mathcal{D}} Q^2, \quad (92)$$

$$\int (1 + |\tau|)^M \|x_\tau\|_2 \|y_\tau\|_2 d|\nu|(\tau) \ll_{M, \mathcal{D}} Q \log(2L). \quad (93)$$

Every component has the same divisor-independent mask, row scales, and no-wrap margins. The applicable TPC-22 static or TPC-23 dynamic mean, discrepancy, and Fourier-tail bounds may be integrated componentwise. For a fixed cell $j \equiv \varrho \pmod{H_0}$, the bounded factor $\Xi_{\alpha, \gamma}(\varrho)$ is absorbed into the row-pair mask. It remains independent of every divisor, additive frequency, and conductor variable.

Proof. This is the continuous log-Mellin separation proved for the same physical provenance in TPC-25. Each fixed smooth row-orbit factor has a Schwartz Mellin transform. Its row phase and orbit phase have modulus one, so the row norms are exactly equations (25) and (26) up to the Schwartz density. Schwartz moments prove equations (92) and (93). Smooth separation does not touch the generic mask.

The constants in the applicable TPC-22 or TPC-23 component estimates grow at most polynomially in finitely many seminorms of Φ_τ , while the weighted moments above are finite. Apply the triangle inequality and Tonelli's theorem to finite Riemann sums, then use dominated convergence. The physical Fourier tail is automatic for this provenance by the rapid-decay argument of TPC-26; the base support $u, v \leq S$ is a subfamily of the polynomial divisor box treated there. This proves the transfer. $\square$

Theorem 6.2 (Physical nonzero centered-base closure). *Assume (89) with fixed margins and all inherited primitive/no-wrap interfaces. Then for every fixed $B > 0$,*

$$\boxed{\mathcal{N}_S^{ZZ, \neq 0} \ll_{B, \mathcal{D}} XQ(\log X)^{-B}.} \quad (94)$$

Proof. Apply lemma 6.1. On every Mellin component, TPC-20's two-scale completion removes the single-indicator and constant nonzero spectra under (90). If $S = R$, apply the TPC-22 static exact-conductor theorem; if $S > R$, apply the TPC-23 dynamic theorem. Their common ledger (88) gives the same fixed-power saving. The relevant Ramanujan-mean theorem bounds the global principal character cell by $Q^2 X^\varepsilon$, a fixed-power saving from $Q^2 J$ because J is a fixed positive power. Integrate the component estimates using (93), the mean using (92), and choose the Fourier-tail decay order after B . Every contribution is then $O_B(XQ(\log X)^{-B})$. $\square$

6.3 The three calibration channels

The fact that $E_S(m)$ depends on the row prevents it from being removed as one global scalar. Uniform pointwise calibration and the actual physical ℓ^1 -mass nevertheless suffice.

Lemma 6.3 (Orbit mass of one centered prefix). *For every retained row and every uniformly bounded physical orbit test supported on an interval of length $O(J)$,*

$$\sum_{j \in \mathbb{Z}} |F(j/J)Z_{m,S}(j)| \ll_{\mathcal{D}} (J+S)(\log(2X))^4. \quad (95)$$

The same bound holds after imposing $(j, H) = 1$ and a fixed residue cell modulo H_0 .

Proof. For $(u, mH) = 1$, the congruence $u \mid mj + h$ selects one class modulo u . Hence its intersection with an interval of length $O(J)$ has $O(J/u + 1)$ points. From (39),

$$\sum_j |F(j/J)Z_{m,S}(j)| \ll \sum_{u \leq S} |b_R(u)| \left(\frac{J}{u} + 1 \right).$$

TPC-24's prefix mass bound gives $\sum_{u \leq S} |b_R(u)|/u \ll (\log(2X))^3$. Pointwise, the defining formula (30) gives

$$|\lambda'_R(u)| \leq \frac{u}{\varphi(u)} \sum_{b \leq R/u} \frac{\mu^2(b)}{\varphi(b)} \ll (\log(2X))^2$$

on its squarefree support; also $|a(u)| \leq \log(2X)$. Hence $\sum_{u \leq S} |b_R(u)| \ll S(\log(2X))^2$. Combining these estimates proves (95). Additional fixed congruence restrictions only reduce the number of orbit points. $\square$

Let $\mathcal{E}_S^{EZ}$, $\mathcal{E}_S^{ZE}$, and $\mathcal{E}_S^{EE}$ denote the three physical terms obtained from the last three summands of (48).

Theorem 6.4 (Scalar calibration closure). *Assume the actual physical provenance and the strict Poisson-short condition (90). Then, for every fixed $B > 0$,*

$$\boxed{|\mathcal{E}_S^{EZ}| + |\mathcal{E}_S^{ZE}| + |\mathcal{E}_S^{EE}| \ll_{B,\mathcal{D}} XQ(\log X)^{-B}.} \quad (96)$$

Proof. Choose the exponent A in (43) after the requested B . Delete the bounded mask, residue factor, and orbit test in absolute value. By equations (25) and (95),

$$|\mathcal{E}_S^{EZ}| + |\mathcal{E}_S^{ZE}| \ll_A Q^2(J+S)(\log(2X))^4(\log X)^{-A}, \quad (97)$$

$$|\mathcal{E}_S^{EE}| \ll_A JQ^2(\log X)^{-2A}. \quad (98)$$

Condition (90) gives $J+S \asymp J$, and $JQ^2 \asymp XQ$. Increase A after the requested B to obtain (96). $\square$

This proof treats EZ, ZE, EE as genuine row-dependent channels. It does not replace them by a single scalar or infer their smallness from the nonzero conductor theorem.

7 Calibrated base closure and passage beyond the Poisson length

We can now state the complete selected-base theorem and splice it to the annular cutoff result of TPC-26. Every conclusion concerns the same physical row family, mask, residue factor, and smooth orbit weight at the two cutoffs.

7.1 The calibrated base theorem

Theorem 7.1 (Physical calibrated base closure). *Retain all hypotheses and definitions of sections 2 and 4, in particular $R \leq S$, the actual opened-row coefficients, and the divisor-independent off-diagonal mask. Assume also the inherited TPC-20–23 primitive, no-wrap, Fourier-tail, and fixed-margin interfaces. Write the scales as in (87). If*

$$s < 1 - \beta, \quad \eta_S(\beta, s) > 0 \quad (99)$$

with fixed margins, then for every fixed $B > 0$,

$$\boxed{\mathcal{Q}_S^{\text{phys}} \ll_{B, \mathcal{D}} XQ(\log X)^{-B}.} \quad (100)$$

Here $\mathcal{Q}_S^{\text{phys}}$ is the calibrated physical packet (46); the theorem is not merely an estimate for its separated nonzero conductor cells.

Proof. Use the exact decomposition (48). For ZZ , first extract physical additive frequency zero before smooth separation and apply theorem 5.3. Then separate only the nonzero spectrum and apply theorem 6.2. The single and constant nonzero channels removed by the TPC-20 two-scale completion, the dynamic Ramanujan mean, and the physical Fourier tail are already included in that theorem’s reassembly. Finally apply theorem 6.4 to EZ, ZE, EE . All fixed dyadic, sign, content, residue, and smooth decompositions cost at most a fixed power of $\log X$, which is absorbed by choosing the calibration and tail orders after B . $\square$

Remark 7.2 (Static base). At $S = R = X^{1/2-\delta+o(1)}$, the proof uses the TPC-22 static theorem for the nonzero component. Its conductor saving is

$$\eta_*(\beta, \delta) = \min \left\{ 1 - \frac{3\beta}{2}, \frac{\beta + 4\delta - 1}{2}, \frac{6\delta - \beta}{4} \right\},$$

which is exactly $\eta_S(\beta, s)$ after $s = 1/2 - \delta$. The additive-zero and scalar bridges in the present paper are still required; the static conductor theorem alone does not display them.

7.2 Splicing to an upper cutoff

Let $T = X^{t+o(1)}$ with $S < T$. TPC-26 proves, under its annular support and symmetric exact-conductor hypotheses, that

$$\mathcal{Q}_T^{\text{phys}} - \mathcal{Q}_S^{\text{phys}} \ll_{B, \mathcal{D}} XQ(\log X)^{-B} \quad (101)$$

whenever its sharp relaxed-box minimax $M_\beta(t)$ is positive with a fixed margin [8]. For $1/2 \leq \beta < 1$,

$$M_\beta(t) = \begin{cases} 1 - \beta - t, & 0 \leq t \leq (1 - \beta)/2, \\ (3 - 3\beta - 2t)/4, & (1 - \beta)/2 \leq t \leq \beta/2, \\ (3 - \beta - 6t)/4, & \beta/2 \leq t \leq (3\beta - 1)/2, \\ (\beta + 1 - 4t)/2, & t \geq (3\beta - 1)/2. \end{cases} \quad (102)$$

Corollary 7.3 (Selected calibrated passage). *Assume the hypotheses of theorem 7.1, with $1/2 \leq \beta < 1$. Suppose also that $R \leq S < T$ lies in the TPC-26 annular interface and*

$$M_\beta(t) > 0 \quad (103)$$

with a fixed margin. Then, for every fixed $B > 0$,

$$\boxed{\mathcal{Q}_T^{\text{phys}} \ll_{B, \mathcal{D}} XQ(\log X)^{-B}.} \quad (104)$$

The upper cutoff T may exceed the Poisson length J . If, in addition, $s < 1 - \beta < t$, this is a genuine passage from a Poisson-short base to a Poisson-long upper cutoff. The row family, both coefficient sequences, mask, periodic residue factor, smooth orbit weight, and prefix definition are the same at the two cutoffs.

Proof. Use the exact identity

$$\mathcal{Q}_T^{\text{phys}} = \mathcal{Q}_S^{\text{phys}} + (\mathcal{Q}_T^{\text{phys}} - \mathcal{Q}_S^{\text{phys}})$$

and combine (100) with (101). $\square$

The conclusion is stronger than cutoff stability alone: the base value is now estimated in the overlap with the inherited Poisson-short region. It remains a selected truncated-square theorem and does not restore the complement beyond T .

7.3 An exact rational passage sample

Take

$$\delta = \frac{3}{20}, \quad \beta = \frac{31}{50}, \quad s = \frac{7}{20}, \quad t = \frac{39}{100}. \quad (105)$$

Then

$$R = S = X^{7/20+o(1)}, \quad J = X^{19/50+o(1)}, \quad T = X^{39/100+o(1)}, \quad (106)$$

so

$$S < J < T. \quad (107)$$

The strict Poisson-short margin is

$$(1 - \beta) - s = \frac{3}{100}. \quad (108)$$

The three base conductor entries are

$$\left(1 - \frac{3\beta}{2}, \frac{\beta + 1 - 4s}{2}, \frac{3 - \beta - 6s}{4}\right) = \left(\frac{7}{100}, \frac{11}{100}, \frac{7}{100}\right), \quad (109)$$

so the centered nonzero conductor ledger has margin $7/100$. The complete calibrated base conclusion remains the arbitrary-logarithmic estimate (100). Since

$$\frac{\beta}{2} = \frac{31}{100} < \frac{39}{100} < \frac{3\beta - 1}{2} = \frac{43}{100}, \quad (110)$$

the third line of (102) applies and gives

$$M_{31/50}(39/100) = \frac{3 - 31/50 - 6(39/100)}{4} = \frac{1}{100}. \quad (111)$$

Thus the annular minimax has margin $1/100$; as usual, the estimates may use any fixed exponent strictly below the displayed ledger margin.

Corollary 7.4 (Closed strict crossing sample). *For the actual opened-row packet with the fixed interfaces above, the parameters (105) satisfy*

$$\mathcal{Q}_T^{\text{phys}} \ll_{B, \emptyset} X Q(\log X)^{-B} \quad (112)$$

for every fixed $B > 0$. This upgrades the corresponding TPC-26 sample from a conditional base splice to a closed selected calibrated truncated square.

7.4 The high- β boundary

TPC-26 also records

$$\beta = \frac{267}{400}, \quad s = \frac{23}{60}, \quad t = \frac{193}{500}. \quad (113)$$

Here

$$s - (1 - \beta) = \frac{23}{60} - \frac{133}{400} = \frac{61}{1200} > 0. \quad (114)$$

Thus the base already lies beyond J , and (99) fails. The annular theorem still proves cutoff stability between S and T , but the present paper does not supply its initial value. No high- β base closure is claimed.

7.5 What remains

After [corollary 7.3](#), three independent walls remain:

- (i) a base nonzero theorem when the lower cutoff itself is beyond the Poisson length, needed for samples such as (113);
- (ii) the ultra-long complement beyond the selected upper cutoff, whose divisor reflection produces polynomial families of two-point affine Möbius correlations;
- (iii) the separate positivity and main-term problem required to pass from an analytic upper/cancellation estimate to a prime-pair asymptotic or lower bound.

The first two are analytic cancellation gates. The third remains parity-sensitive. None is resolved by the calibrated base theorem.

8 Exact certificate, logical boundaries, and next gate

This final section separates finite algebra, imported analytic input, new analytic output, and statements that remain open.

8.1 Reproducible finite certificate

The standard-library script

```
experiments/tpc27_certificate.py
```

uses exact rational arithmetic and formal prime-log variables. It checks the Ramanujan compatibility resolution, the truncated and complete finite-model multiple identities, the formal prime-multiple identity, the Ramanujan-square zero factorization, the exact primitive orbit bridge $P = Z + E$, the four-channel algebra, and both rational exponent ledgers. It contains no floating-point tolerance and uses an explicit exception-based checker, so optimization cannot delete a test.

Both

```
python experiments/tpc27_certificate.py,
python -O experiments/tpc27_certificate.py
```

complete 20,540 exact checks and produce the same JSON file. The recorded hashes are

Source SHA-256	59afa0a8f94bc208297478d1fcc41e85 6cf88351c03bae3ef0e51690676a8e3d
JSON SHA-256	dd7033c1a170a6ba968d4bc478841e19 978a5cbb38401a9c7d64681889bd2a48
Certificate digest	ae5b89520111cf8be04cc193e85cd197 b8159cbcd1b9e0421b9d8f2a1af0e6c4

The certificate controls finite algebra only; it is not asymptotic or prime-pair evidence.

8.2 Dependency and status ledger

Item	Status	Source or limitation
Ramanujan-square zero factorization	exact, unconditional	Theorem 3.3 ; finite sums only
Finite-model multiple density	exact, unconditional	Proposition 4.1 ; completeness uses $kdH \leq R$
Uniform prime-multiple calibration	proved imported input	Classical zero-free region in the polynomial modulus range used by TPC-20 and TPC-24
Centered physical base zero	closed here	Off-diagonal mask, small/large k split, and actual row ℓ^1 -mass
Centered physical base nonzero spectrum	closed by transfer	TPC-22 when $S = R$, TPC-23 when $S > R$, with the TPC-25–26 continuous physical transfer
Scalar channels EZ, ZE, EE	closed here	Uniform endpoint calibration and the strict Poisson-short orbit bound
Sample (3/20, 31/50, 7/20, 39/100)	fully closed selected square	Centered nonzero conductor margin 7/100, annular margin 1/100, and arbitrary logarithmic saving in the calibrated conclusion
High- β sample of TPC-26	base open	Its lower cutoff already satisfies $S/J = X^{61/1200+o(1)}$
Ultra-long complement and full residual	open	Requires new cancellation for affine two-point Möbius families
Positivity or Hardy–Littlewood main term	open	Not implied by a selected cancellation bound

8.3 Why the interfaces cannot be silently dropped

Three finite examples isolate the logical boundaries. First, if a bilinear form is $\mathfrak{B}(x, y) = xy$, taking $Z_1 = Z_2 = 0$ and $E_1 = E_2 = 1$ makes the centered square vanish but the physical square equal one. Thus a centered estimate does not restore the scalar channels. Second, on an even orbit the two legs $C(j) = (-1)^j/2$ each have mean zero, while the additive zero mode of $C(j)^2$ is $1/4$. Separate centering does not remove the joint zero frequency. Third, on the row diagonal, $c_k(0) = \varphi(k)$, so the large-modulus K_0^{-1} tail used in [theorem 5.1](#) disappears. These examples explain, respectively, the explicit EZ, ZE, EE restoration, zero extraction before separation, and the off-diagonal mask.

8.4 Next analytic gate

The next step has two branches: prove a centered nonzero base theorem for $S > J$, or isolate a verifiable affine two-point Möbius subfamily in the reflected complement. Failure can yield a precise barrier theorem; success would still require a separate positivity and main-term mechanism before twin primes.

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